Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) "Coding Requirements" section:

FR-DLT-001: Load Customer and Order Data into Delta Tables

Title: Load Customer and Order Data into Delta Tables

Description: The system must read source CSV data from a specified volume and load it into Delta tables for customer and order data.

Preconditions:

CSV files are available at the specified locations: `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata` and `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`.

- The Delta tables for customer and order data do not exist or are empty.

Main Flow / Functional Steps:

- Read customer CSV data from `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata` into a Delta table named "customer".
- Read order CSV data from `/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata` into a Delta table named "order".

The schema for the "customer" table is: CustId, Name, EmailId, Region.

The schema for the "order" table is: OrderId, ItemName, PricePerUnit, Qty, Date, CustId.

FR-DLT-002: Transform Order Data and Remove Null/Duplicate Records

Title: Transform Order Data and Remove Null/Duplicate Records

Description: The system must add a new column "TotalAmount" to the "order" table, remove null and duplicate records from both "customer" and "order" tables.

Preconditions:

- The "customer" and "order" Delta tables exist and contain data.

Main Flow / Functional Steps:

- Add a new column "TotalAmount" to the "order" table by multiplying "PricePerUnit" and "Qty" column values.
- Remove null records from both "customer" and "order" tables.

- Remove duplicate records from both "customer" and "order" tables.

FR-DLT-003: Create ordersummary Table and Load Joined Data

Title: Create ordersummary Table and Load Joined Data

Description: The system must create an "ordersummary" table if it does not exist and load joined data from "customer" and "order" tables into it using SCD type 2.

Preconditions:

- The "customer" and "order" Delta tables exist and contain data.
- The "ordersummary" table does not exist or is empty.

Main Flow / Functional Steps:

- Create the "ordersummary" table in catalog="gen_ai_poc_databrickscoe" and schema="sdlc_wizard" if it does not exist.
- Join "customer" and "order" data using the "CustId" field.
- Load the joined data into the "ordersummary" table using SCD type 2.

FR-DLT-004: Update ordersummary Table on Customer Data Change

Title: Update ordersummary Table on Customer Data Change

Description: The system must update the "ordersummary" table whenever there is a change in the "customer" table.

Preconditions:

- The "customer" and "ordersummary" tables exist and contain data.

Main Flow / Functional Steps:

- Detect changes in the "customer" table.
- Update the "ordersummary" table by making old records inactive and new records active.
- Update the "StartDate" and "EndDate" columns accordingly to maintain history.

FR-DLT-005: Create customeraggregatespend Table and Load Aggregated Data

Title: Create customeraggregatespend Table and Load Aggregated Data

Description: The system must create a "customeraggregatespend" table if it does not exist and load aggregated data from the "ordersummary" table into it.

Preconditions:

- The "ordersummary" table exists and contains data.
- The "customeraggregatespend" table does not exist or is empty.

Main Flow / Functional Steps:

- Create the "customeraggregatespend" table in catalog="gen_ai_poc_databrickscoe" and schema="sdlc_wizard" if it does not exist.
- Aggregate the "TotalAmount" column from the "ordersummary" table by grouping by "Name" and "Date" columns.
- Load the aggregated data into the "customeraggregatespend" table.

FR-DLT-006: Implement using Delta Live Tables

Title: Implement using Delta Live Tables

Description: The system must implement the above requirements using Delta Live Tables.

Preconditions:

- All previous requirements have been implemented.

Main Flow / Functional Steps:

- Implement the data pipeline using Delta Live Tables to perform the required data transformations and loading.